Chapter 1 Representation of Compact Groups

1.1 Basic Representation Theory

1.1.1 Definitions

The Topology of E

Representation of Topological Group

Definition 1.1.1.1 Let G be a topological group, and E be a topological vector space.

- We say that: E is a G-module, if there is an action

$$G \times E \to E \qquad (g, x) \mapsto gx$$

- an action $(g, x) \mapsto gx : G \times E \to E$ such that:
  - $: x \mapsto gx, E \to E$ is a **continuous vector space endomorphism** of E, for each $g \in G$;
  - $g \mapsto gx, G \to E$ is continuous for each $x \in E$;
- A **representation** of G is a continuous map

$$\pi : G \to \mathcal{L}_p(E)$$

satysfying

$$\pi(1) = id_E \qquad \pi(gh) = \pi(g)\pi(h); g, h \in G$$

Remark

- Possible Representation

$$\pi : G \to \mathcal{L}_p(E) \qquad g \mapsto (x \mapsto gx)$$

- If $\pi : G \to \mathcal{L}_p(E)$ is a representation, then the function

$$G \times E \to E \qquad (g, x) \mapsto gx := \pi(g)(x)$$

endows E with the structure of a G-module.

We could have another famous statement: It is the theory of G-module.

▶ ***Theorem*** Assume that: E is a G-module for a topological group G, and that

$$\pi : G \to \mathcal{L}_p(E)$$

is the associated representation.
If: E is a Baire space, then for every compact subspace K of G, the set

$$\pi(K) \subseteq Hom(E, E)$$

is equi-continuous at 0. That is: For any neighborhood V of 0 in E, there is a neighborhood U of 0 such that

$$KU \subseteq V$$

As a consequence, if G is locally compact, the function

$$G \times E \to E \qquad (g, x) \mapsto gx$$

is continuous.

◦ **Proof** ... □

♣ **Example 1.1.1.2** ...
★ **Corollary 1.1.1.3** ...

◦ **Proof** ... □

1.1.2 Haar Integral

Take the following notion: Given a measure μ,

$$\mu(f) := \int_X f d\mu = (\mu, f)$$

or indeed $= \int_G f(g) d\mu(g)$. A special case is the classical Fourier analysis

$$\mathbb{T} = \mathbb{R}/\mathbb{Z}$$

Invariant Measure

Take the following notation $({}_g f)(x) := f(g^{-1}x); g \in G$(multiplicative group).

Definition 1.1.2.1

- Let G denote a compact group.
  A measure μ is called **invariant** if

$$\mu({}_g f) = \mu(f), \forall g \in G; f \in E = C(G, \mathbb{K})$$

- A measure μ is called **Haar measure**, if it is invariant and positive.
  Positive: $\mu(f) \geq 0, \forall f \geq 0$.
- The measure is **normalized** if $\mu(1) = 1$.

♣ **Example 1.1.2.2** Consider the morphism

$$p : \mathbb{R} \to \mathbb{T} \qquad t \mapsto t + \mathbb{Z}$$

♣ **Example 1.1.2.3** ...

Measure on Group

Preparations for existence theorem:
◇ **Lemma 1.1.2.4** ...

◦ **Proof** ... □

▶ ***Theorem*** (The existence and Uniqueness Theorem of Haar Measure) For each **compact group**

∘ **Proof** ... □

★ **Corollary 1.1.2.5** ...

∘ **Proof** ... □

1.1.3 Consequence of Haar Measure

▶ ***Theorem*** (Weyl's Trick)

- Let G be a compact group, E a G-module which is also a Hilbert space.(not Hilbert G-module) Then, there is a scalar product relative to which all operators $\pi(g)$ are unitary.
- Specifically, if $(-,-)$ is the given scalar product on E, then

$$< x|y >= \int_G (gx|gy)dg$$

defines a scalar product such that

$$M^{-2}(x|x) \leq \langle x|x \rangle \leq M^2(x|x) \qquad M = sup\{\sqrt{(gx|gx)}|g \in G, (x|x) \leq 1\}$$

and that

$$\langle gx|gy \rangle = \langle x|y \rangle; x, y \in E, g \in G$$

∘ **Proof** Firstly, the new-defined product is well-defined. Then, since Haar measure is positive, we have $(gx|gx) \geq 0 \Rightarrow \langle x|x \rangle \geq 0$. Thus, the constant M is well-defined.

Then,

...

Finally:

$$< hx|hy >= \int_G (ghx|ghy)dg = \int_G (gx|gy)dg =< x, y > \qquad \text{(Invariance of Haar Measure)}$$

□

Hilbert G-module

Definition 1.1.3.1 (Hilbert G-module) If G is a topological space. Then, a **Hilbert G-module** is a Hilbert space E and a G-module such that: All operators $\pi(g)$ is unitary. i.e.

$$(gx|gy) = (x, y); x, y \in E, g \in G$$

♣ **Example 1.1.3.2** Let G be a compact group and $\mathcal{H}_0$ the vector space $C(G, \mathcal{K})$ equipped with the scalar product

$$(f_1|f_2) = \gamma(f_1\bar{f_2}) = \int_G f_1(g)\overline{f_2(g)}dg$$

Indeed:

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Then, the open set $U = \{t \in G|(f\bar{f})(t) > 0\}$ contains g. Hence is nonempty.

1.1.4 The Main Theorem on Hilbert Modules for Compact Groups

Recall:

- A **sesquilinear form** is a function

  $$B : \mathcal{H} \times \mathcal{H} \to K$$

  that is linear in the first and conjugate linear in the second. (in the sense of $\mathbb{C}$-linear)
- It is **bounded** if

  $$|B(x, y)| \leq M\|x\| \cdot \|y\| \qquad x, y \in \mathcal{H}$$

- A way to define sesquilinear form: Suppose T be a **bounded linear operator** on $\mathcal{H}$. Then,

  $$B(x, y) = (Tx|y)$$

  defines a sesquilineaar from with $M = \|T\|$: In the view of Cauchy-Schwartz

  $$|(x, y)| \leq \|x\| \cdot \|y\|$$

But conversely:

◇ **Lemma 1.1.4.1** If B is a sesquilinear form, then there exists a unique bounded operator T of $\mathcal{H}$ such that

$$\|T\| \leq M \qquad B(x, y) = (Tx|y)$$

∘ **Proof** Such T is specified by:
Uniqueness: □

Under this trick, we could define another operator, which is stabil under group actions:

◇ **Lemma 1.1.4.2** Let G denote a compact grooup and T a bounded operator on a Hilbert G-module E. Then: There exists a unique bounded operator $\tilde{T}$ on E, with $\|\tilde{T}\| \leq \|T\|$ such that

$$(\tilde{T}x|y) = \int_G (Tgx|gy)dg = \int_G (\pi(g)^{-1}T\pi(g)(x)|y)dg$$

∘ **Proof** ... □

♠ **Proposition 1.1.4.3** (Criterion for $\tilde{S}$) The following statements are equivalent for an operator S of H:

- $S \in Hom_G(H, H)$;

- $S = \tilde{S}$;
- There is an operator T such that $S = \tilde{T}$;

∘ **Proof** ... □

G-**Submodules**

Definition 1.1.4.4 If G is a topological group, E a G-module, then a vector-subspace V of E is called a **submodule** if

$$GV \subseteq V$$

also called **invariant subspace**.

◇ **Lemma 1.1.4.5** ...

∘ **Proof** ... □

◇ **Lemma 1.1.4.6** ...

∘ **Proof** ... □

1.1.5 The Fundamental Theorem on Unitary Modules

1.1.6 Definition of a Compact Lie Group,Compact Group by Compact Lie Groups

Given in Lie Theory.

1.2 Theorem of Peter and Weyl

1.3 G-Module Theory

1.3.1 Vector Valued Integration

Chapter 2 Characters

Chapter 3 Lie Group

These will be given in the note of Lie group and Algebras;

- Linear Lie Groups;
- Compact Lie Groups;

Chapter 4 Abelian Groups